

SESSION-8:

Tasks

1.

Declare a function called `getUserInitials` that takes a user's full name (like 'Virat Kohli') and returns their initials in uppercase (e.g., 'VK'). Call this function with your favorite cricketer's name and print the result.

```
#include <stdio.h>
```

```
#include <ctype.h>
```

```
void getUserInitials(char name[], char initials[]) {  
    int j = 0;  
  
    for (int i = 0; name[i] != '\0'; i++) {  
        if (i == 0 || (name[i - 1] == ' ' && name[i] != ' ')) {  
            initials[j++] = toupper(name[i]);  
        }  
    }  
  
    initials[j] = '\0';  
}
```

```
int main() {  
    char name[] = "Rushi Patel";  
    char initials[10];  
  
    getUserInitials(name, initials);
```

```
printf("Initials: %s\n", initials);
```

```
return 0;
```

```
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session8_task1.c -o session8_task1
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session8_task1.exe
Initials: RP
```

2.

Create a function `addToCart` that takes a shopping cart array and a product name, adds the product to the cart, and prints the updated cart. Demonstrate how passing the cart array by reference allows changes to persist outside the function.

Hint: In languages like JavaScript, arrays are passed by reference. In C/C++, use pointers for reference behavior.

```
#include <stdio.h>
```

```
#include <string.h>
```

```
void addToCart(char cart[][50], int *cartSize, char product[]) {
```

```
    strcpy(cart[*cartSize], product);
```

```
    (*cartSize)++;
```

```
    printf("\nUpdated Cart:\n");
```

```
    for (int i = 0; i < *cartSize; i++) {
```

```
        printf("%d. %s\n", i + 1, cart[i]);
```

```
    }
```

```
}
```

```
int main() {
```

```
    char cart[10][50] = {
```

```

        "Laptop",
        "Headphones"
    };

    int cartSize = 2;

    printf("Original Cart:\n");
    for (int i = 0; i < cartSize; i++) {
        printf("%d. %s\n", i + 1, cart[i]);
    }

    addToCart(cart, &cartSize, "Smartphone");

    return 0;
}

```

```

PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session8_task2.c -o session8_task2
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session8_task2.exe
Original Cart:
1. Laptop
2. Headphones

Updated Cart:
1. Laptop
2. Headphones
3. Smartphone

```

3.

Write two functions: `increaseFollowersByValue` and `increaseFollowersByReference`. Each should take a followers count and add 1000 to it, but one should use pass-by-value and the other pass-by-reference. Show how the original followers count changes (or doesn't) after calling each function.

```
#include <stdio.h>
```

```
void increaseFollowersByValue(int followers) {  
    followers += 1000;  
    printf("Inside pass-by-value: %d\n", followers);  
}
```

```
void increaseFollowersByReference(int *followers) {  
    *followers += 1000;  
    printf("Inside pass-by-reference: %d\n", *followers);  
}
```

```
int main() {  
    int followers = 5000;  
  
    printf("Original followers: %d\n", followers);  
  
    increaseFollowersByValue(followers);  
    printf("After pass-by-value: %d\n", followers);  
  
    increaseFollowersByReference(&followers);  
    printf("After pass-by-reference: %d\n", followers);  
  
    return 0;
```

```

}
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session8_task3.c -o session8_task3
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session8_task3.exe
Original followers: 5000
Inside pass-by-value: 6000
After pass-by-value: 5000
Inside pass-by-reference: 6000
After pass-by-reference: 6000

```

4.

Build a function `formatPrice` that takes a price in rupees and returns a string formatted like Flipkart's price tag (e.g., '₹1,599'). Use this function to display prices for three different products.

```
#include <stdio.h>
```

```

void formatPrice(int price) {
    if (price >= 1000) {
        printf("₹%d,%03d\n", price / 1000, price % 1000);
    } else {
        printf("₹%d\n", price);
    }
}

```

```

int main() {
    int price1 = 1599;
    int price2 = 24999;
    int price3 = 799;

    printf("Product 1: ");
    formatPrice(price1);

    printf("Product 2: ");
    formatPrice(price2);

```

```

printf("Product 3: ");
formatPrice(price3);

return 0;
}

```

```

PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session8_task4.c -o session8_task4
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session8_task4.exe
Product 1: ₹1,599
Product 2: ₹24,999
Product 3: ₹799

```

5.

Refactor an existing function you wrote above to make it reusable for both product names and usernames (for example, a function that capitalizes the first letter of any string).

Constraint: The refactored function should work for any string input, not just a specific use case.

```

#include <stdio.h>
#include <ctype.h>

void capitalizeFirstLetter(char text[]) {
    if (text[0] != '\0') {
        text[0] = toupper(text[0]);
    }
}

int main() {
    char productName[] = "laptop";
    char username[] = "siddhi";

    capitalizeFirstLetter(productName);
    capitalizeFirstLetter(username);

    printf("Product Name: %s\n", productName);
    printf("Username: %s\n", username);
}

```

```
    return 0;  
}
```

```
PS C:\Users\SIDDHI\OneDrive\Desktop\c> gcc session8_task5.c -o session8_task5  
PS C:\Users\SIDDHI\OneDrive\Desktop\c> .\session8_task5.exe  
Product Name: Laptop  
Username: RUSHI
```